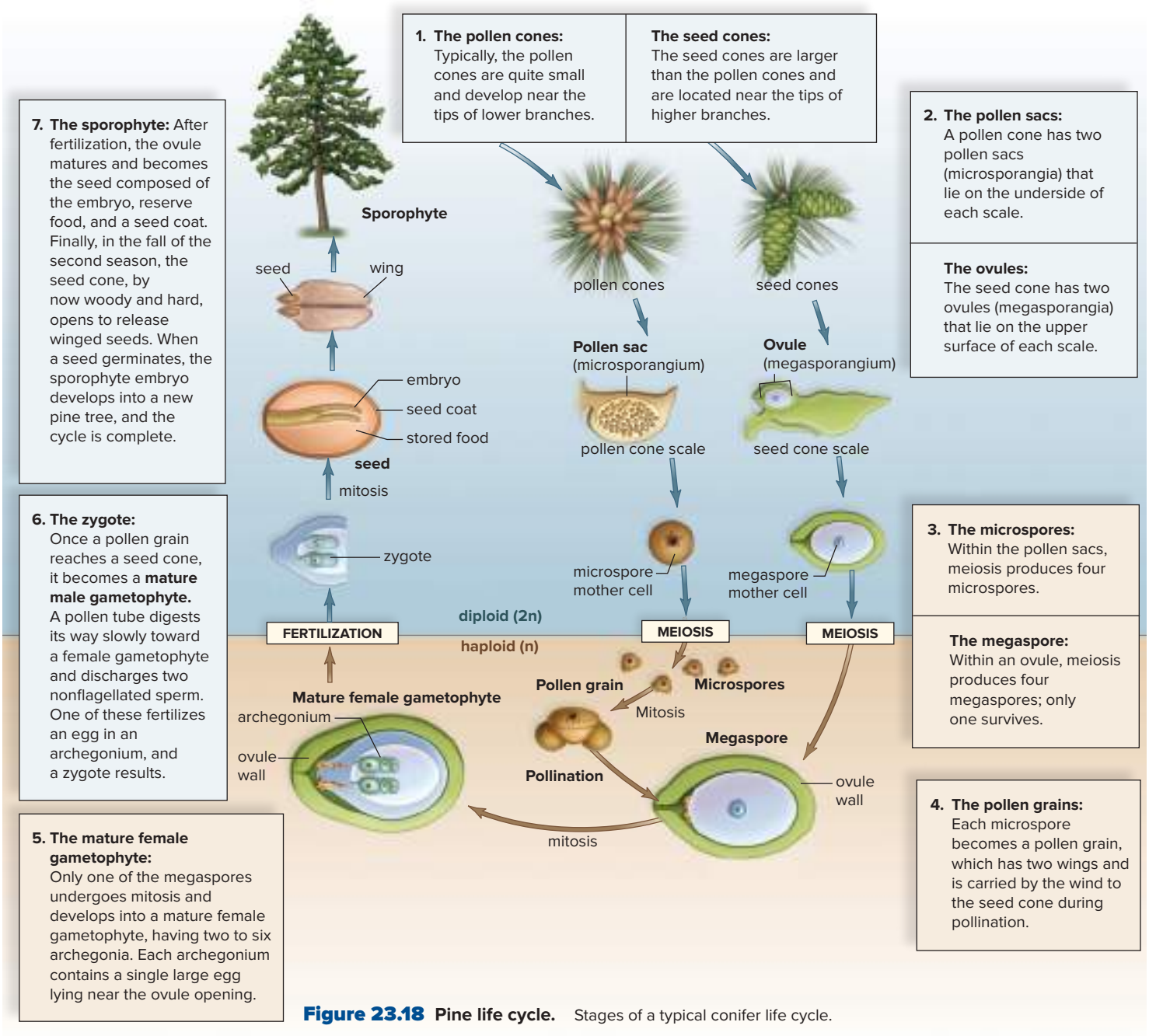


Figure 23.18 Pine life cycle. Stages of a typical conifer life cycle.

recessed stomata. Note that in the life cycle of the pine (Fig. 23.18) the sporophyte is dominant, pollen grains are windblown, and the seed is what is dispersed. Conifers are **monoecious,** which means that a single plant carries both male and female reproductive structures.

Pines, in particular, are well-known for their beauty and pleasant smell; they make attractive additions to parks and gardens. Pines have medicinal value in that the needles and bark are rich in vitamins A and C and can be used to create teas that ease symptoms of a cold or cough. Large pine seeds, called pine nuts, are sometimes harvested for use in cooking, and pine oil is used to scent a number of household and personal care products, such as room sprays and deodorant.

Cycads

Cycads include ten genera and 300 species of distinctive gymnosperms. The cycads are native to tropical and subtropical forests. *Zamia pumila,* found in Florida, is the only species of cycad native to North America. Cycads are commonly used in landscaping. One species, *Cycas revoluta,* referred to as the sago palm, is a common landscaping plant. Their large, finely divided leaves grow in clusters at the top of the stem, and therefore they resemble palms or ferns, depending on their height. The trunk of a cycad is unbranched, even if it reaches a height of 15–18 m, as is possible in some species.

Unlike conifers, cycads have pollen and seed cones on separate plants. The cones, which grow at the top of the stem surrounded by the leaves, can be huge—more than a meter long with a weight of



a. An African cycad



b. *Ginkgo biloba,* a native of China

Figure 23.19 Three groups of gymnosperms. **a.** Cycads may resemble ferns or palms, but they are cone-producing gymnosperms. **b.** A ginkgo tree has broad leaves and fleshy seeds borne at the end of stalklike megasporophylls. **c.** *Ephedra,* a type of gnetophyte, is a branched shrub. This specimen produces pollen in microsporangia. **d.** *Welwitschia,* another type of gnetophyte, produces two straplike leaves that fray and split. (a): (cycad): Hoberman Collection/Alamy Stock Photo; (cycad cone): PlazaCameraman/iStock/Getty Images; (b): (Ginkgo leaves): blickwinkel/Koenig/Alamy Stock Photo; (ovule): Ken Robertson, University of Illinois/INHS; (c): (plant): Evelyn Jo Johnson/McGraw Hill; (microsporangia): Karl J. Niklas; (d): TravelMuse/Alamy Stock Photo



c. *Ephedra*, a type of gnetophyte



d. *Welwitschia*, a type of gnetophyte

40 kg (Fig. 23.19*a*). Cycads exhibit the life cycle of a gymnosperm, except they are pollinated by insects rather than by wind. Also, the pollen tube bursts in the vicinity of the archegonium, and multiflagellated sperm swim to reach an egg.

Cycads were plentiful in the Mesozoic era at the time of the dinosaurs, and it's likely that dinosaurs fed on cycad seeds. Now, cycads are in danger of extinction, because they grow very slowly.

Ginkgoes

Although **ginkgoes** are plentiful in the fossil record, they are represented today by only one surviving species of tree, the *Ginkgo biloba*. Ginkgoes are **dioecious,** which means that a single plant produces either male or female reproductive structures, but not both (Fig. 23.19*b*). The fleshy seeds, which ripen in the fall, give off such a foul odor that male trees are usually preferred for planting. Ginkgo trees are resistant to pollution and do well along city streets and in city parks. Ginkgo is native to China, and in Asia, ginkgo seeds are considered a delicacy. Extracts from ginkgo trees have been used to improve blood circulation.

Like cycads, the pollen tube of ginkgo bursts to release multiflagellated sperm that swim to the egg produced by the female gametophyte, located within an ovule.

Gnetophytes

Gnetophytes are represented by three living genera and 95 species of plants that are very diverse in appearance. In all gnetophytes, xylem is structured similarly, none have archegonia, and their strobili (cones) have a similar construction. The reproductive structures of some gnetophyte species produce nectar, and insects play a role in the pollination of these species.

Gnetum, which occurs in the tropics, consists of trees or climbing vines with broad, leathery leaves arranged in pairs. *Ephedra,* occurring only in southwestern North America and Southeast Asia, is a shrub with small, scalelike leaves (Fig. 23.19*c*). Ephedrine, a medicine with serious side effects, is extracted from *Ephedra. Welwitschia,* living in the deserts of southwestern Africa, has only two enormous, straplike leaves that fray as it gets older (Fig. 23.19*d*).

Angiosperms

Angiosperms (Gk. *angion,* "vessel"; *sperma,* "seed") are the flowering plants. They are an exceptionally large and successful group, with roughly 370,000 species thought to exist—six times the number of all other plant groups combined. Angiosperms live in all sorts of habitats, from fresh water to desert and from the frigid north to the torrid tropics. They range in size from the tiny, almost microscopic duckweed to *Eucalyptus* trees over 100 m tall.

It would be impossible to exaggerate the importance of angiosperms in our everyday lives. Angiosperms include all the hardwood trees of temperate deciduous forests and all the broadleaved evergreen trees of tropical forests. Also, all herbaceous (nonwoody

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